

KITABU QUEST

S1

Physics

Mascot: mountain gorilla

Scene: learners exploring physics with simple apparatus and a mountain gorilla mascot

CBC aligned draft learner book

Think • Explore • Achieve

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Title Page

KITABU QUEST S1 Physics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S1 learners study Physics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use measurements, diagrams, models, and step-by-step reasoning about systems and forces.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Laboratory safety rules and measurements of physical quantities
2. Qualitative analysis of linear motion
3. Force (I)
4. Newton's Laws of Motion (I)
5. Centre of gravity
6. Work, Power and Energy(i)
7. Simple Machines (I)
8. Kinetic theory and states of matter
9. Heat and temperature
10. Magnetism (I)
11. Electrostatic (I)
12. Current Electricity (I)

As you finish each topic, return to the success criteria and correct one answer before moving on.

Laboratory safety rules and measurements of physical quantities: Lesson Opener

Learning focus: Laboratory safety rules and measurements of physical quantities

Country link: a safe Rwandan observation, model, data table, or investigation for Laboratory safety rules and measurements of physical quantities.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain the importance of physics, measure physical quantities and express findings in appropriate units

Inquiry questions:

- How can laboratory safety rules and measurements of physical quantities help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Laboratory safety rules and measurements of physical quantities: Learn

Laboratory safety rules and measurements of physical quantities begins with safe behaviour and accurate measurement.

Core idea:

- Physics uses measurements such as length, mass, time, temperature, and volume.
- A measurement is useful only when the number and unit are both written.
- Safety rules protect learners, equipment, and results.
- Read scales at eye level and record values immediately.

Local link: in Rwanda, careful measurement helps in health, building, farming, weather records, transport, and school laboratory work.

Key words:

- Laboratory safety rules and measurements of physical quantities
- Laboratory
- safety
- rules
- measurements
- physical
- quantities
- concept

Laboratory safety rules and measurements of physical quantities: Worked Example

Investigation model for Laboratory safety rules and measurements of physical quantities: Measure the length of a desk safely.

1. Choose a metre rule or tape measure.
2. Place zero at one end of the desk.
3. Read the scale at eye level.
4. Record the value with the unit, for example 120 cm.
5. Repeat once and compare readings.

Conclusion: a useful measurement includes a number, a unit, and a careful method.

Laboratory safety rules and measurements of physical quantities: Vocabulary And Study Skill

Use these words and study moves before you practise Laboratory safety rules and measurements of physical quantities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Laboratory safety rules and measurements of physical quantities
- Laboratory
- safety
- rules
- measurements
- physical
- quantities
- concept

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Laboratory safety rules and measurements of physical quantities without a clear example, method, or evidence.

Laboratory safety rules and measurements of physical quantities: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Laboratory safety rules and measurements of physical quantities.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Laboratory safety rules and measurements of physical quantities: Practice And Correction

Practice for Laboratory safety rules and measurements of physical quantities:

Main outcome: explain the importance of physics, measure physical quantities and express findings in appropriate units.

1. Name three laboratory safety rules.
2. Measure one classroom object and record the value with a unit.
3. Explain one error that can make a measurement unreliable.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Laboratory safety rules and measurements of physical quantities: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Laboratory safety rules and measurements of physical quantities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Laboratory safety rules and measurements of physical quantities: Outcome Check

Outcome check for Laboratory safety rules and measurements of physical quantities: Explain the importance of physics, measure physical quantities and express findings in appropriate units.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Qualitative analysis of linear motion: Lesson Opener

Learning focus: Qualitative analysis of linear motion

Country link: a safe Rwandan observation, model, data table, or investigation for Qualitative analysis of linear motion.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- describe objects in motion in one dimension using the principles of kinematics

Inquiry questions:

- How can qualitative analysis of linear motion help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Qualitative analysis of linear motion: Learn

Qualitative analysis of linear motion explains how position changes with time.

Core idea:

- Motion means an object changes position compared with a reference point.
- Distance tells how far an object moves.
- Time tells how long the motion takes.
- Speed compares distance and time.
- A distance-time table or graph can show whether motion is slow, fast, or changing.

Key words:

- Qualitative analysis of linear motion
- Qualitative
- analysis
- linear
- motion
- concept
- observation
- evidence

Qualitative analysis of linear motion: Worked Example

Worked example for Qualitative analysis of linear motion: A cart moves 12 metres in 4 seconds. Find its speed.

1. Speed = distance / time.
2. Distance = 12 m and time = 4 s.
3. $12 / 4 = 3$.

Answer: the cart moves at 3 m/s.

Qualitative analysis of linear motion: Vocabulary And Study Skill

Use these words and study moves before you practise Qualitative analysis of linear motion. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Qualitative analysis of linear motion
- Qualitative
- analysis
- linear
- motion
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Qualitative analysis of linear motion without a clear example, method, or evidence.

Qualitative analysis of linear motion: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Qualitative analysis of linear motion.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Qualitative analysis of linear motion: Practice And Correction

Practice for Qualitative analysis of linear motion:

Main outcome: describe objects in motion in one dimension using the principles of kinematics.

1. A bicycle moves 30 m in 6 s. Calculate its speed.
2. Draw a two-row table for distance and time.
3. Explain what happens to speed if the same distance takes more time.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Qualitative analysis of linear motion: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Qualitative analysis of linear motion.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Qualitative analysis of linear motion: Outcome Check

Outcome check for Qualitative analysis of linear motion: Describe objects in motion in one dimension using the principles of kinematics.

1. Calculate the speed of an object that moves 20 m in 5 s.
2. State the formula used.
3. Write the unit.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Force (I): Lesson Opener

Learning focus: Force (I)

Country link: a safe Rwandan observation, model, data table, or investigation for Force (I).

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- define, explain, and describe forces and their effects

Inquiry questions:

- How can force (i) help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Force (I): Learn

Force (I) explains pushes, pulls, and changes in motion.

Core idea:

- A force can start motion, stop motion, change speed, change direction, or change shape.
- Forces are measured in newtons.
- Balanced forces do not change motion.
- Unbalanced forces can make an object accelerate, slow down, or turn.

Key words:

- Force (I)
- Force
- concept
- observation
- evidence
- safety
- conclusion

Force (I): Worked Example

Worked example for Force (I): A learner pushes a box, and the box starts moving.

Observation: the push is an unbalanced force.

Explanation:

1. The box was at rest.
2. The push acted on the box.
3. The box changed its motion, so the force had an effect.

Conclusion: forces can change the motion of objects.

Force (I): Vocabulary And Study Skill

Use these words and study moves before you practise Force (I). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Force (I)
- Force
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Force (I) without a clear example, method, or evidence.

Force (I): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Force (I).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Force (I): Practice And Correction

Practice for Force (I):

Main outcome: define, explain, and describe forces and their effects.

1. Name two pushes and two pulls from daily life.
2. State one effect of force on a moving object.
3. Draw arrows to show forces on a box being pushed.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Force (I): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Force (I).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Force (I): Outcome Check

Outcome check for Force (I): Define, explain, and describe forces and their effects.

1. Name one push and one pull.
2. State one effect of an unbalanced force.
3. Draw an arrow to show force direction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Newton's Laws of Motion (I): Lesson Opener

Learning focus: Newton's Laws of Motion (I)

Country link: a safe Rwandan observation, model, data table, or investigation for Newton's Laws of Motion (I).

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- state Newton's laws to describe the effects of forces on objects

Inquiry questions:

- How can Newton's laws of motion (i) help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Newton's Laws of Motion (I): Learn

Newton's Laws of Motion (I) explains how forces affect motion using Newton's laws.

Core idea:

- First law: an object stays at rest or keeps moving in a straight line unless an unbalanced force acts on it.
- Second law: a larger force causes a larger acceleration, while a larger mass is harder to accelerate.
- Third law: when one object exerts a force on another, the second object exerts an equal and opposite force.
- Force diagrams help you show direction and balance.

Key words:

- Newton's Laws of Motion (I)
- Newton's
- Laws
- Motion
- concept
- observation
- evidence
- safety

Newton's Laws of Motion (I): Worked Example

Worked example for Newton's Laws of Motion (I): Explain why a football at rest starts moving when kicked.

1. Before the kick, the ball is at rest.
2. The foot applies an unbalanced force on the ball.
3. Because of Newton's first law, the ball changes its state of motion only when the force acts.
4. The ball also pushes back on the foot with an equal and opposite force, showing Newton's third law.

Conclusion: the kick changes the motion because it provides an unbalanced force.

Newton's Laws of Motion (I): Vocabulary And Study Skill

Use these words and study moves before you practise Newton's Laws of Motion (I). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Newton's Laws of Motion (I)
- Newton's
- Laws
- Motion
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Newton's Laws of Motion (I) without a clear example, method, or evidence.

Newton's Laws of Motion (I): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Newton's Laws of Motion (I).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Newton's Laws of Motion (I): Practice And Correction

Practice for Newton's Laws of Motion (I):

Main outcome: state Newton's laws to describe the effects of forces on objects.

1. State Newton's first, second, and third laws in your own words.
2. Draw a force diagram for a pushed box or kicked ball.
3. Explain which law is shown when the object starts moving, speeds up, or pushes back.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Newton's Laws of Motion (I): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Newton's Laws of Motion (I).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Newton's Laws of Motion (I): Outcome Check

Outcome check for Newton's Laws of Motion (I): State Newton's laws to describe the effects of forces on objects.

1. State one Newton law correctly.
2. Apply it to a moving or resting object.
3. Draw or describe the force direction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Centre of gravity: Lesson Opener

Learning focus: Centre of gravity

Country link: a safe Rwandan observation, model, data table, or investigation for Centre of gravity.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- determine the position of centre of gravity of a body

Inquiry questions:

- How can centre of gravity help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Centre of gravity: Learn

Centre of gravity explains balance and stability.

Core idea:

- Centre of gravity is the point where the weight of an object seems to act.
- A low centre of gravity usually makes an object more stable.
- A wide base of support helps an object resist toppling.
- You can investigate balance using safe objects such as rulers, cardboard shapes, and books.

Key words:

- Centre of gravity
- Centre
- gravity
- concept
- observation
- evidence
- safety
- conclusion

Centre of gravity: Worked Example

Investigation model for Centre of gravity: Balance a ruler on one finger.

1. Place a ruler across one finger.
2. Move your finger slowly until the ruler balances.
3. Mark the balance point.
4. Add a small rubber on one side and observe the new balance point.

Conclusion: adding mass changes where the centre of gravity acts.

Centre of gravity: Vocabulary And Study Skill

Use these words and study moves before you practise Centre of gravity. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Centre of gravity
- Centre
- gravity
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Centre of gravity without a clear example, method, or evidence.

Centre of gravity: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Centre of gravity.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Centre of gravity: Practice And Correction

Practice for Centre of gravity:

Main outcome: determine the position of centre of gravity of a body.

1. Balance a safe flat object and mark the balance point.
2. Explain why a wide base improves stability.
3. Name one object designed to avoid toppling.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Centre of gravity: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Centre of gravity.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Centre of gravity: Outcome Check

Outcome check for Centre of gravity: Determine the position of centre of gravity of a body.

1. Explain why a low, wide object is more stable.
2. Name one example.
3. Use the term centre of gravity correctly.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Work, Power and Energy(i): Lesson Opener

Learning focus: Work, Power and Energy(i)

Country link: a safe Rwandan observation, model, data table, or investigation for Work, Power and Energy(i).

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- analyse the process of energy transformations and conservation

Inquiry questions:

- How can work, power and energy(i) help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Work, Power and Energy(i): Learn

Work, Power and Energy(i) connects work, power, and energy changes.

Core idea:

- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:

- Work, Power and Energy(i)
- Work
- Power
- Energy(i)
- concept
- observation
- evidence
- safety

Work, Power and Energy(i): Worked Example

Worked example for Work, Power and Energy(i): A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Work, Power and Energy(i): Vocabulary And Study Skill

Use these words and study moves before you practise Work, Power and Energy(i). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Work, Power and Energy(i)
- Work
- Power
- Energy(i)
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Work, Power and Energy(i) without a clear example, method, or evidence.

Work, Power and Energy(i): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Work, Power and Energy(i).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Work, Power and Energy(i): Practice And Correction

Practice for Work, Power and Energy(i):

Main outcome: analyse the process of energy transformations and conservation.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Work, Power and Energy(i): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Work, Power and Energy(i).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Work, Power and Energy(i): Outcome Check

Outcome check for Work, Power and Energy(i): Analyse the process of energy transformations and conservation.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Simple Machines (I): Lesson Opener

Learning focus: Simple Machines (I)

Country link: a safe Rwandan observation, model, data table, or investigation for Simple Machines (I).

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- analyse relationship among energy, work and power for simple machines

Inquiry questions:

- How can simple machines (i) help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Simple Machines (I): Learn

Simple Machines (I) shows how tools make work easier.

Core idea:

- A simple machine changes the size or direction of a force.
- Levers, pulleys, wheels and axles, inclined planes, wedges, and screws are simple machines.
- A machine does not remove work; it helps you use force more conveniently.
- Compare effort force, load, and distance moved.

Key words:

- Simple Machines (I)
- Simple
- Machines
- concept
- observation
- evidence
- safety
- conclusion

Simple Machines (I): Worked Example

Investigation model for Simple Machines (I): Use a ruler as a lever.

1. Place a small object as the load.
2. Put a pencil under the ruler as the pivot.
3. Press the other end gently.
4. Move the pivot and compare the effort needed.

Conclusion: the position of the pivot changes how a lever helps lift a load.

Simple Machines (I): Vocabulary And Study Skill

Use these words and study moves before you practise Simple Machines (I). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Simple Machines (I)
- Simple
- Machines
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Simple Machines (I) without a clear example, method, or evidence.

Simple Machines (I): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Simple Machines (I).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Simple Machines (I): Practice And Correction

Practice for Simple Machines (I):

Main outcome: analyse relationship among energy, work and power for simple machines.

1. Name three simple machines.
2. Identify the load, effort, and pivot in a lever.
3. Explain how changing the pivot changes the effort.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Simple Machines (I): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Simple Machines (I).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Simple Machines (I): Outcome Check

Outcome check for Simple Machines (I): Analyse relationship among energy, work and power for simple machines.

1. Name one simple machine.
2. Identify load, effort, and pivot or contact point.
3. Explain how it makes work easier.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Kinetic theory and states of matter: Lesson Opener

Learning focus: Kinetic theory and states of matter

Country link: a safe Rwandan observation, model, data table, or investigation for Kinetic theory and states of matter.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- relate physical properties of solids, liquids and gases to temperature

Inquiry questions:

- How can kinetic theory and states of matter help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Kinetic theory and states of matter: Learn

Kinetic theory and states of matter explains particles in solids, liquids, and gases.

Core idea:

- Matter is made of tiny particles.
- In solids, particles are close and vibrate in fixed positions.
- In liquids, particles are close but can slide past each other.
- In gases, particles are far apart and move freely.
- Heating usually makes particles move faster.

Key words:

- Kinetic theory and states of matter
- Kinetic
- theory
- states
- matter
- concept
- observation
- evidence

Kinetic theory and states of matter: Worked Example

Model for Kinetic theory and states of matter: Compare particles in solids, liquids, and gases.

1. Draw solid particles close together in a fixed pattern.
2. Draw liquid particles close but uneven, able to slide.
3. Draw gas particles far apart.
4. Add arrows to show particle movement.

Conclusion: particle arrangement explains shape, flow, and expansion.

Kinetic theory and states of matter: Vocabulary And Study Skill

Use these words and study moves before you practise Kinetic theory and states of matter. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Kinetic theory and states of matter
- Kinetic
- theory
- states
- matter
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Kinetic theory and states of matter without a clear example, method, or evidence.

Kinetic theory and states of matter: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Kinetic theory and states of matter.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Kinetic theory and states of matter: Practice And Correction

Practice for Kinetic theory and states of matter:

Main outcome: relate physical properties of solids, liquids and gases to temperature.

1. Draw particles in a solid, liquid, and gas.
2. Explain why gases fill containers.
3. State what heating does to particle movement.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Kinetic theory and states of matter: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Kinetic theory and states of matter.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Kinetic theory and states of matter: Outcome Check

Outcome check for Kinetic theory and states of matter: Relate physical properties of solids, liquids and gases to temperature.

1. Compare particles in a solid and a gas.
2. Draw a simple particle diagram.
3. Explain what heating changes.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Heat and temperature: Lesson Opener

Learning focus: Heat and temperature

Country link: a safe Rwandan observation, model, data table, or investigation for Heat and temperature.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain principle of thermometry and compare different temperature scales

Inquiry questions:

- How can heat and temperature help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Heat and temperature: Learn

Heat and temperature explains thermal energy and temperature.

Core idea:

- Temperature tells how hot or cold something is.
- Heat moves from a hotter object to a cooler object.
- Thermometers must be read carefully and safely.
- Different scales, such as Celsius and Kelvin, can describe temperature.

Key words:

- Heat and temperature
- Heat
- temperature
- concept
- observation
- evidence
- safety
- conclusion

Heat and temperature: Worked Example

Investigation model for Heat and temperature: Read water temperature safely.

1. Put warm water in a beaker under teacher guidance.
2. Place a thermometer bulb in the water without touching the bottom.
3. Wait for the reading to stop changing.
4. Read the Celsius scale at eye level.

Conclusion: careful thermometer use gives a reliable temperature.

Heat and temperature: Vocabulary And Study Skill

Use these words and study moves before you practise Heat and temperature. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Heat and temperature
- Heat
- temperature
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Heat and temperature without a clear example, method, or evidence.

Heat and temperature: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Heat and temperature.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Heat and temperature: Practice And Correction

Practice for Heat and temperature:

Main outcome: explain principle of thermometry and compare different temperature scales.

1. Read a thermometer diagram or classroom thermometer safely.
2. State the direction heat flows between hot and cold objects.
3. Compare heat and temperature in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Heat and temperature: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Heat and temperature.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Heat and temperature: Outcome Check

Outcome check for Heat and temperature: Explain principle of thermometry and compare different temperature scales.

1. State the direction heat flows.
2. Describe safe thermometer reading.
3. Use the word temperature correctly.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetism (I): Lesson Opener

Learning focus: Magnetism (I)

Country link: a safe Rwandan observation, model, data table, or investigation for Magnetism (I).

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- differentiate magnetic and non-magnetic materials

Inquiry questions:

- How can magnetism (i) help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetism (I): Learn

Magnetism (I) explains magnetic and non-magnetic materials.

Core idea:

- Magnets attract some materials, especially iron and steel.
- A magnet has north and south poles.
- Like poles repel and unlike poles attract.
- Magnetic force can act without touching the object.

Key words:

- Magnetism (I)
- Magnetism
- concept
- observation
- evidence
- safety
- conclusion

Magnetism (I): Worked Example

Investigation model for Magnetism (I): Test materials with a magnet.

1. Collect a nail, coin, plastic ruler, paper clip, and wood piece.
2. Bring a magnet near each object.
3. Record attracted or not attracted.
4. Group the materials as magnetic or non-magnetic.

Conclusion: iron and steel objects are usually attracted by magnets.

Magnetism (I): Vocabulary And Study Skill

Use these words and study moves before you practise Magnetism (I). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Magnetism (I)
- Magnetism
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Magnetism (I) without a clear example, method, or evidence.

Magnetism (I): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Magnetism (I).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Magnetism (I): Practice And Correction

Practice for Magnetism (I):

Main outcome: differentiate magnetic and non-magnetic materials.

1. Test five safe materials with a magnet.
2. Record attracted/not attracted in a table.
3. State what happens when like poles are brought together.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Magnetism (I): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Magnetism (I).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetism (I): Outcome Check

Outcome check for Magnetism (I): Differentiate magnetic and non-magnetic materials.

1. Name two magnetic materials.
2. State what happens with like poles.
3. Describe a fair magnet test.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrostatic (I): Lesson Opener

Learning focus: Electrostatic (I)

Country link: a safe Rwandan observation, model, data table, or investigation for Electrostatic (I).

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain charging of materials and distribution of electric charges on conductors

Inquiry questions:

- How can electrostatic (i) help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrostatic (I): Learn

Electrostatic (I) explains electric charges at rest.

Core idea:

- Some materials become charged when rubbed.
- Charged objects can attract or repel other charged objects.
- Like charges repel; unlike charges attract.
- Dry materials often show static effects more clearly than wet materials.

Key words:

- Electrostatic (I)
- Electrostatic
- concept
- observation
- evidence
- safety
- conclusion

Electrostatic (I): Worked Example

Investigation model for Electrostatic (I): Observe static attraction.

1. Rub a plastic ruler with dry cloth.
2. Bring it near small paper pieces.
3. Record what happens without touching the paper first.
4. Repeat after waiting one minute.

Conclusion: rubbing can charge materials and cause attraction.

Electrostatic (I): Vocabulary And Study Skill

Use these words and study moves before you practise Electrostatic (I). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electrostatic (I)
- Electrostatic
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electrostatic (I) without a clear example, method, or evidence.

Electrostatic (I): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Electrostatic (I).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electrostatic (I): Practice And Correction

Practice for Electrostatic (I):

Main outcome: explain charging of materials and distribution of electric charges on conductors.

1. Describe how to charge a plastic ruler safely.
2. State whether like charges attract or repel.
3. Name one situation where static charge is noticed.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electrostatic (I): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electrostatic (I).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrostatic (I): Outcome Check

Outcome check for Electrostatic (I): Explain charging of materials and distribution of electric charges on conductors.

1. Describe how rubbing can charge an object.
2. State whether like charges attract or repel.
3. Give one safe classroom observation.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Current Electricity (I): Lesson Opener

Learning focus: Current Electricity (I)

Country link: a safe Rwandan observation, model, data table, or investigation for Current Electricity (I).

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain different effects of electricity and safety precautions to observe while using electricity

Inquiry questions:

- How can current electricity (i) help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Current Electricity (I): Learn

Current Electricity (I) explains moving electric charge in circuits.

Core idea:

- A complete circuit is needed for current to flow.
- Cells, wires, switches, and bulbs can show current effects safely.
- Electricity can produce light, heat, sound, movement, and magnetism.
- Safety matters: never experiment with mains electricity.

Key words:

- Current Electricity (I)
- Current
- Electricity
- concept
- observation
- evidence
- safety
- conclusion

Current Electricity (I): Worked Example

Circuit model for Current Electricity (I): Light a bulb safely.

1. Use a cell, two wires, a switch, and a small bulb.
2. Connect a complete circuit.
3. Close the switch and observe the bulb.
4. Open the switch and observe again.

Conclusion: current flows only when the circuit is complete.

Current Electricity (I): Vocabulary And Study Skill

Use these words and study moves before you practise Current Electricity (I). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Current Electricity (I)
- Current
- Electricity
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Current Electricity (I) without a clear example, method, or evidence.

Current Electricity (I): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Current Electricity (I).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Current Electricity (I): Practice And Correction

Practice for Current Electricity (I):

Main outcome: explain different effects of electricity and safety precautions to observe while using electricity.

1. Draw a complete circuit with a cell, switch, and bulb.
2. Explain why a broken circuit stops current.
3. List two electrical safety rules.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Current Electricity (I): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Current Electricity (I).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Current Electricity (I): Outcome Check

Outcome check for Current Electricity (I): Explain different effects of electricity and safety precautions to observe while using electricity.

1. Draw a complete circuit.
2. Explain why a switch can stop current.
3. Write one electrical safety rule.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Rectilinear Propagation of light: Lesson Opener

Learning focus: Rectilinear Propagation of light

Country link: a safe Rwandan observation, model, data table, or investigation for Rectilinear Propagation of light.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain the nature of light, rectilinear propagation of light and reflection at plane surface

Inquiry questions:

- How can rectilinear propagation of light help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Rectilinear Propagation of light: Learn

Rectilinear Propagation of light explains how light travels and reflects.

Core idea:

- Light travels in straight lines through a uniform medium.
- Shadows form when an opaque object blocks light.
- Reflection occurs when light bounces from a surface.
- Smooth, shiny surfaces reflect light more regularly than rough surfaces.

Key words:

- Rectilinear Propagation of light
- Rectilinear
- Propagation
- light
- concept
- observation
- evidence
- safety

Rectilinear Propagation of light: Worked Example

Investigation model for Rectilinear Propagation of light: Show that light travels in straight lines.

1. Make small holes in three cards.
2. Place a torch behind the first card.
3. Align the holes so light reaches a screen.
4. Move one card sideways and observe the screen.

Conclusion: the light path is blocked when the holes are not in a straight line.

Rectilinear Propagation of light: Vocabulary And Study Skill

Use these words and study moves before you practise Rectilinear Propagation of light. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Rectilinear Propagation of light
- Rectilinear
- Propagation
- light
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Rectilinear Propagation of light without a clear example, method, or evidence.

Rectilinear Propagation of light: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Rectilinear Propagation of light.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Rectilinear Propagation of light: Practice And Correction

Practice for Rectilinear Propagation of light:

Main outcome: explain the nature of light, rectilinear propagation of light and reflection at plane surface.

1. Draw a straight ray of light from a torch.
2. Explain how a shadow forms.
3. Name one smooth reflector and one rough surface.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Rectilinear Propagation of light: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Rectilinear Propagation of light.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Rectilinear Propagation of light: Outcome Check

Outcome check for Rectilinear Propagation of light: Explain the nature of light, rectilinear propagation of light and reflection at plane surface.

1. Draw a straight light ray.
2. Explain how a shadow forms.
3. Describe reflection from a smooth surface.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Laboratory safety rules and measurements of physical quantities: Review Clinic 1

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Laboratory safety rules and measurements of physical quantities that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Qualitative analysis of linear motion: Review Clinic 2

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Qualitative analysis of linear motion that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Force (I): Review Clinic 3

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Force (I) that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Newton's Laws of Motion (I): Review Clinic 4

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Newton's Laws of Motion (I) that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Physics.

- Laboratory safety rules and measurements of physical quantities: Explain this term using an example from the book, your school, or your community.
- Qualitative analysis of linear motion: Explain this term using an example from the book, your school, or your community.
- Force (I): Explain this term using an example from the book, your school, or your community.
- Newton's Laws of Motion (I): Explain this term using an example from the book, your school, or your community.
- Centre of gravity: Explain this term using an example from the book, your school, or your community.
- Work, Power and Energy(i): Explain this term using an example from the book, your school, or your community.
- Simple Machines (I): Explain this term using an example from the book, your school, or your community.
- Kinetic theory and states of matter: Explain this term using an example from the book, your school, or your community.
- Heat and temperature: Explain this term using an example from the book, your school, or your community.
- Magnetism (I): Explain this term using an example from the book, your school, or your community.
- Electrostatic (I): Explain this term using an example from the book, your school, or your community.
- Current Electricity (I): Explain this term using an example from the book, your school, or your community.
- Rectilinear Propagation of light: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Physics answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Physics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.